

Delivery Route Optimizer

Team Composition:

Tejarsha Chappidi (Individual Project)

Problem

In many real-world delivery systems, a delivery agent needs to deliver packages to multiple locations starting from a warehouse or distribution center. When the delivery route is not planned properly, the agent may travel unnecessary distances, which increases fuel consumption, travel time, and operational cost.

As the number of delivery locations increases, manually deciding the best order to visit each location becomes difficult. The delivery agent may end up taking longer routes or revisiting areas unnecessarily.

The objective of this project is to design a system that determines efficient routes between multiple locations such that the total travel distance is minimized. By optimizing the delivery route, the system can help reduce travel time and improve efficiency.

This problem is relevant to real-world applications such as courier delivery, food delivery services, e-commerce logistics, and transportation planning systems.

Dataset

For this project, we will create a custom dataset that represents a simplified delivery network.

The dataset will include:

- a warehouse location that acts as the starting point
- multiple delivery locations that need to be visited
- distances between locations that represent travel cost

Each row in the dataset will represent a connection between two locations and the distance between them. The distance value represents how far the delivery agent needs to travel between those two points.

The dataset simulates a real-world road network where different locations are connected through routes. Using this dataset, we can represent the delivery system as a graph structure and apply graph algorithms to determine efficient paths.

Example format of dataset:

Source	Destination	Distance
Warehouse	A	5
Warehouse	B	7
A	C	3
B	C	2
C	D	4

The dataset will be prepared during the implementation phase and will include enough locations to demonstrate the working of the algorithm.

Algorithm and its Usage

We will represent the delivery network as a weighted graph.

In this graph:

- each location (warehouse or delivery point) will be represented as a node
- each route between two locations will be represented as an edge
- the distance between locations will be represented as the weight of the edge

To determine the most efficient route, we will use **Dijkstra's Algorithm**, which computes the shortest path between nodes in a weighted graph.

Dijkstra's Algorithm starts from a source node (warehouse) and calculates the minimum distance to all other nodes by continuously selecting the path with the smallest cumulative distance. The algorithm ensures that the computed path between locations results in the least possible travel distance.

This approach focuses on computing shortest paths between delivery locations starting from the warehouse. Using these results, the system will suggest an efficient route order that helps the delivery agent minimize total travel distance. The algorithm ensures that the chosen path reduces unnecessary travel compared to random or manually selected routes.

This demonstrates how graph algorithms can be applied to solve practical optimization problems in logistics and transportation systems.

AI Usage Statement

AI tools were used only to assist with formatting and improving the clarity of the written content. No AI tools were used to develop the core algorithm, logic, or approach of this project. All ideas and problem formulation were independently developed.

